CMS · NANOAOD · TRIGGER OBJECTS

The unseeded diphoton leg never gets its NanoAOD filter bits

Photon **TrigObj_filterBits** 16 and 17 (14 and 15 before NanoAOD v15) are documented for HLT_Diphoton30_22_R9Id_OR_IsoCaloId_AND_HE_R9Id_Mass90 / Mass95, but no photon trigger object ever carries them. Why it happens, how widespread it is, and the one-line fix.

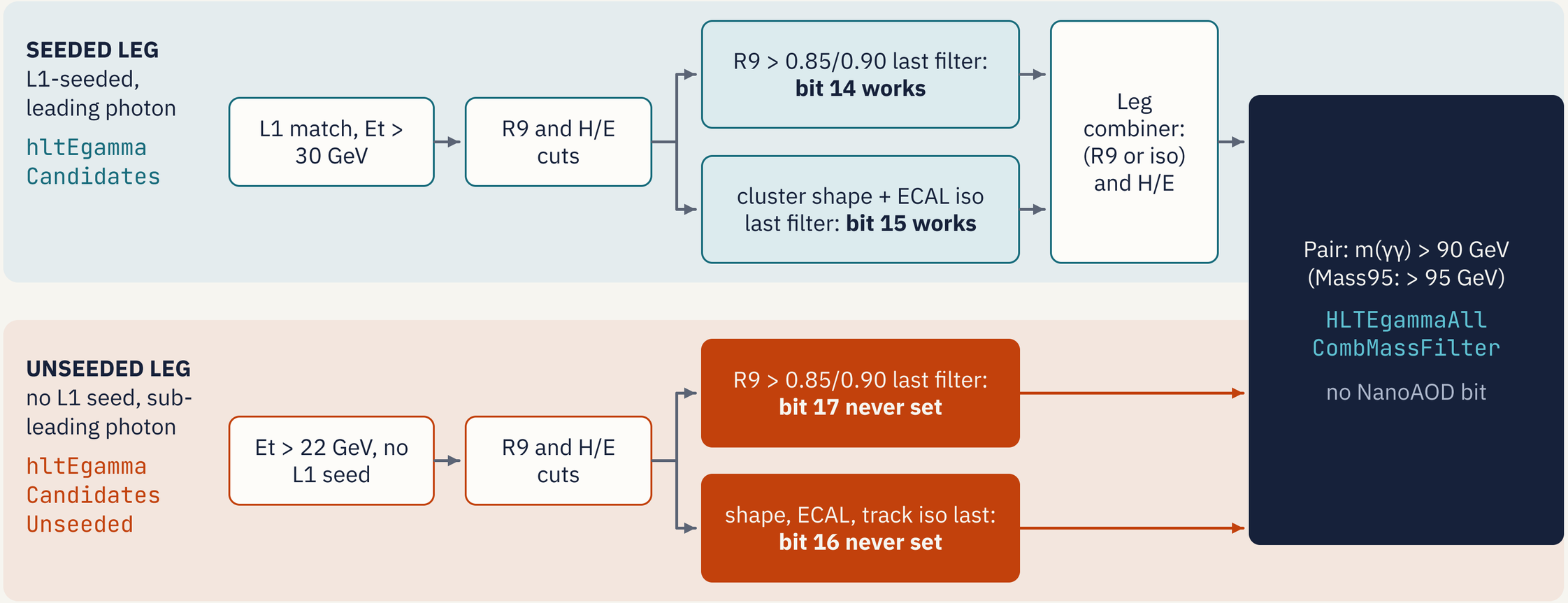
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Checked against cms-sw/cmssw master ed35654b1 (today), release branches 13_0_X to 16_1_X, and the 2018 to 2026 HLT menus

`PhysicsTools/NanoAOD/python/triggerObjects_cff.py`

One path, two legs, four last filters

HLT_Diphoton30_22_R9Id_OR_IsoCaloId_AND_HE_R9Id_Mass90 and Mass95 (same sequence), every menu 2018 to 2026



Orange boxes: the unseeded last filters NanoAOD discards. Bits in v15 numbering; v12 to v14: 12, 13, 15, 14.

From the HLT filter to TrigObj_filterBits: the object survives four hand-offs and dies at the fifth

`1 · HLT filters (saveTags)`	The unseeded last filters tag their objects with the collection **hltEgammaCandidatesUnseeded**; the leg's first filter saves them with type 92, the later ones with type 81.
↓	
`2 · hltTriggerSummaryAOD`	Objects are stored once per HLT collection; every filter that accepted an object records its key and the type it saved it with.
↓	
`3 · patTrigger → slimmedPatTrigger (MiniAOD)`	The pat object keeps the collection label **hltEgammaCandidatesUnseeded::HLT**, all its types (92 and 81) and the labels of every filter it passed.
↓	
`4 · unpackedPatTrigger (NanoAOD)`	Filter labels are unpacked; the object reaches the table producer intact, with its collection name and types.
↓	
`5 · triggerObjectTable, Photon selection`	**sel = type(92) && pt > 15 && coll('hltEgammaCandidates')** is evaluated before the quality bits. The collection test fails, the object is dropped, and bits 16 and 17 are never computed.

Where it breaks: the producer's decision for one unseeded photon object

1 **type(92)?** yes. The leg's first filter (HLT1Photon, triggerType 92) saved the object as TriggerCluster; PAT keeps every type.

2 **pt > 15?** yes. The unseeded leg requires Et > 22 GeV (18 GeV for the Diphoton30_18 paths).

3 **coll('hltEgammaCandidates')?** no. The stored label is hltEgammaCandidatesUnseeded::HLT. Without a wildcard the comparison is exact (TriggerObjectStandAlone::hasCollection falls back to TriggerObject::hasCollection).

4 **Discarded before the bits.** qualityBits is only evaluated for objects that pass sel; the duplicate merging only sees accepted objects. Bits 16 and 17 stay 0 in every event.

```
# triggerObjects_cff.py, Photon
sel = "type(92) && pt > 15 &&
       coll('hltEgammaCandidates')"
skipObjectsNotPassingQualityBits = True
```

```
// TriggerObjectTableProducer.cc, produce()
for (const auto &obj : trigObjs)
  for (const auto &sel : sels_)
    if (sel.match(obj)) { // fails here
      bits = sel.qualityBits(obj);
      if (bits =/= 0) selected.push_back(...);
    }
```

Same logic in 13_0_X, 13_3_X, 14_0_X, 15_0_X and master; only the bit width (32 bits up to 15_0_1, 64 bits from 15_0_2) and the merging rule changed.

The photon bit list: what is documented versus what can be set

bit v15	bit v12 to v14	filter named by the bit	objects come from	status
8	8	`hltHtEcal800`	hltHtMhtEcal (HT object, type 89)	**never set**: not an e/gamma object, since v10
14	12	`hltEG30LR9Id85b90eHE12R9Id50b80eR9IdLastFilter`	hltEgammaCandidates	**works** (seeded leg)
15	13	`hltEG30LIso60CaloId15b35eHE12R9Id50b80eEcalIsoLastFilter`	hltEgammaCandidates	**works** (seeded leg)
16	14	`hltEG22Iso60CaloId15b35eHE12R9Id50b80eTrackIsoUnseededLastFilter`	hltEgammaCandidatesUnseeded	**never set**: collection rejected by sel
17	15	`hltEG22R9Id85b90eHE12R9Id50b80eR9UnseededLastFilter`	hltEgammaCandidatesUnseeded	**never set**: collection rejected by sel
18	16	`hltEG30Iso60CaloId15b35eR9Id50b90e`	hltEgammaCandidates	**works for 2018 and 2022 menus**; filter

How widespread (1): every NanoAOD version since v12, every menu since 2018, still there today

NanoAOD	release checked	photon selection	dead unseeded bits	affected
v9	CMSSW_10_6_X	seeded only, no photon bits at all	none defined	no photon TrigObj
v10, v11	CMSSW_12_4_X, 12_6_X	seeded only, pt > 20	bits absent	no
v12	CMSSW_13_0_0 to 13_0_17 (prod 13_0_13)	seeded only	14, 15, 17	yes
v13	CMSSW_13_3_1_patch1	seeded only	14, 15, 17	yes
v14	CMSSW_14_0_X, 14_1_X, 14_2_X	seeded only	14, 15, 17	yes
v15	CMSSW_15_0_2 to 15_0_15_patch4, 15_1_X	seeded only	16, 17, 19	yes
next	16_0_X, 16_1_X (tags to 16_1_3), master ed35654b1 today	seeded only, no open PR or issue	16, 17, 19	yes

Menus

19 menu files checked, 2018 GRun to the frozen 2026 menu: the four Diphoton30_22 filters exist everywhere, always on the same collections, with the same 16-filter chain. Path versions v13 (2018, early 2022) to v29

Datasets

Every Run 3 NanoAOD file: 2022 to 2025 data (PromptNano, 22Sep2023, 16Dec2023, v15 re-NANO) and the Summer22, 22EE, 23, 23BPix and 24 MC campaigns, plus any Run 2 input re-processed with v12 or later

Analyses

Any NanoAOD-based matching of the sub-leading photon finds nothing: HiggsDNA-style workflows, the HYgammaRunIII legend that lists bits 16 and 17 as the legs, FLAF configurations. MiniAOD-based flashgg reads

How widespread (2): every filter bit of every object type, resolved against six menus

object	bits	never settable (collection or type)	stale name in the 2025 menu	fine in 2025
Electron	20	none	2: WPLoose (gone 2023), triple-electron (gone 2025)	18
Photon	21	4: bits 16, 17, 19 (unseeded) and 8 (hltHtEcal800)	3: bit 12 (gone 2022), 18 and 19 (renamed 2023)	15
Muon	14	none	2: mu-tau DeepTau overlap filters (gone 2025)	12
Tau	31	none	1: DeepTau (gone 2025)	30
BoostedTau	3	v12 to v14: whole selection, see below	1: 2022 ParticleNetTauTau name	2
Jet	46	none	10: pixel-only HH4b filters (gone 2024), DeepCSV and DeepJet (gone 2025), 3-jet pt40 and pt45	34 (+2 L1 seeds)
FatJet	7	none	3: 2022 and 2023 soft-drop and ParticleNet names	4
HT, MHT	6, 2	none	2: Run 2 L1 seed names (never in Run 3 menus)	3 (+1), 2
MET	0	no bits with the skip flag on: MET objects never written	not applicable	none, all versions

Same mechanism elsewhere

Only in the Photon list: the three unseeded diphoton bits and bit 8, whose filter saves HT objects. Electron and Muon have no collection

Stale names are the common case

One bit list serves the 2022 to 2026 menus, so bits for filters that were renamed or retired stay documented but silently empty for later years,

The fix: one line, mirroring what the Electron selection has done since 2023

```
PhysicsTools/NanoAOD/python/triggerObjects_cff.py, Photon = cms.PSet( ... )
- sel = cms.string("type(92) && pt > 15 && coll('hltEgammaCandidates')"),
+ sel = cms.string("type(92) && pt > 15 && (coll('hltEgammaCandidates') ||
coll('hltEgammaCandidatesUnseeded'))"),
```

Why this form

It names exactly the two e/gamma collections and copies the Electron selection (PR 41827). A wildcard coll('hltEgammaCandidates*') would also match, but would silently admit any future collection with that prefix.

What changes in the output

Unseeded objects enter the photon branch. With skipObjectsNotPassingQualityBits on, only those that set a photon bit are written, and the only photon-bit filters on the unseeded collection are the three diphoton ones. A seeded twin in the same direction is merged with OR-ed bits; otherwise a second photon entry appears, as for electrons.

What does not change

Bit numbers, documentation strings, the other object types, the event content. Existing files cannot be repaired: the objects were never written. Only a new NANO pass with the fix restores the sub-leading leg.

Status. Commit 1699a39f7 on the fork branch claude/fervent-lamport-3oiuh6 and the same change as aebf3eb64 on top of upstream master ed35654b1 (branch nano-photon-trigobj-unseeded). Applies cleanly to today's master. Nothing pushed, no pull request opened yet.

Validation, pull request, and the follow-ups this fix does not cover

Validation (needs a CMSSW area)

- Run standard workflows with a NANO step before and after: H to gamma gamma 12452.0 and 12852.0, TTbar 12434.0 and 12834.0, data 141.046 (RunEGamma2023D); confirm the numbers with runTheMatrix.py -n -e.
- Count photon TrigObj with bits 16 and 17 in events firing the diphoton path: zero before, of the order of the number of triggered events after.
- Only diphoton-triggered events move: nTrigObj and the TrigObj DQM histograms for id 22, nothing else.

Pull request

- Push nano-photon-trigobj-unseeded to the fork, open it against cms-sw:master citing PR 40711 and PR 41827; xpog signature.
- Request a CMSSW_15_0_X backport if another NanoAOD production from that branch is planned.

Follow-ups, separate from the one-line fix

- Add the 2023 names of the Diphoton30_18 filters as alternatives inside bits 18 and 19, so the numbering stays stable.
- Move or drop bit 8 (hltHtEcal800): its objects are HT objects and can never be photons.
- Review the stale bits per year (WPLoose, triple-electron, DeepTau overlaps, pixel-only jet filters, DeepCSV and DeepJet) and say in the documentation which menus each bit covers.
- Bits for the low-mass eta1p5, DiphotonMVA14p25, Diphoton15_1x EBEB and DoublePhoton paths, with the EGM POG and the Higgs PAG.
- Inform XPOG that MET trigger objects are never written and that the BoostedTau table was empty in v12 to v14.

Notes, scripts, patch and PR text: the nanoaod-trigobj-notes repository (bundle delivered). Nothing has been pushed to any GitHub repository.